

Flight Price Prediction Model

This project is a Machine Learning based Flight Price Prediction system that predicts airline ticket prices using various flight-related features. The complete workflow including data preprocessing, feature engineering, model training, and evaluation is implemented in a Jupyter Notebook. The trained model is saved as a .pkl file for reuse.

Project Files

- flight_price_updated.ipynb
Contains the full machine learning pipeline including:
 - Data loading
 - Data cleaning
 - Feature engineering
 - Exploratory Data Analysis (EDA)
 - Model training
 - Model evaluation
 - flight_price_model.pkl
Serialized trained machine learning model saved using Pickle.
This file can be loaded directly to make predictions without retraining the model.
 - README.md
Project documentation.
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Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Pickle
- Jupyter Notebook

Problem Statement

Flight ticket prices vary depending on multiple factors such as airline, source, destination, number of stops, journey date, and duration. The objective of this project is to build a regression model that can accurately predict flight prices based on these input features.

Workflow

1. Data Collection and Loading
 2. Data Cleaning and Handling Missing Values
 3. Feature Engineering
 4. Exploratory Data Analysis
 5. Model Training using Regression Algorithms
 6. Model Evaluation using performance metrics
 7. Model Serialization using Pickle
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How to Run the Project

Step 1: Clone the Repository

```
git clone https://github.com/your-username/flight-price-prediction.git
cd flight-price-prediction
```

Step 2: Install Required Libraries

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

Step 3: Run the Notebook

```
jupyter notebook
```

Open:

```
flight_price_updated.ipynb
```

Using the Saved Model

You can load the trained model from the .pkl file and use it for prediction as shown below:

```
import pickle

with open("flight_price_model.pkl", "rb") as file:
    model = pickle.load(file)

# Example prediction (replace with actual feature values)
prediction = model.predict([[...input_features...]])
print(prediction)
```

Model Evaluation

The model performance is evaluated using:

- R2 Score
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

You can update this section with your actual evaluation scores from the notebook.

Future Enhancements

- Hyperparameter tuning
 - Deployment using Flask or Streamlit
 - Integration with real-time flight APIs
 - Improved feature selection and preprocessing
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